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Lambertian scattering model for nanospheres and nanoshells

In this set of notes, we use the Lambertian scattering model to determine the geometric mean free path in nanospheres and nanoshells. For nanospheres, we derive **Equation 17** of [Electronic properties of small silver particles: the optical constants and their temperature dependence](https://iopscience.iop.org/article/10.1088/0305-4608/4/Z/007/pdf?casa_token=hvvOpv3ti7gAAAAA:VESHyB-K4EUXsxZ6hnH44l4mj3ZqxjTRagpPPfYtxOuLEYSqXhJLpvQZf9D6qftTuLbeiiECLaOBkA77pHZ9_kmfHdc)

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[K4EUXsxZ6hnH44l4mj3ZqxjTRagpPPfYtxOuLEYSqXhJLpvQZf9D6qftTuLbeiiECLaOBkA77pHZ9_kmfHdc](https://iopscience.iop.org/article/10.1088/0305-4608/4/Z/007/pdf?casa_token=hvvOpv3ti7gAAAAA:VESHyB-K4EUXsxZ6hnH44l4mj3ZqxjTRagpPPfYtxOuLEYSqXhJLpvQZf9D6qftTuLbeiiECLaOBkA77pHZ9_kmfHdc)) by

Kreibig (henceforth referred to as **Paper 1**) and for nanospheres we derive

Equation 14 of [Electron Mean Free Path in a Spherical Shell Geometry](http://www.wave-scattering.com/jpchc_112_10641.pdf)

(http://www.wave-scattering.com/jpchc_112_10641.pdf) by **Moroz** (henceforth referred to as

Paper 2).

Section 1: Lambertian surface scattering model for spheres: Deriving Equation 17 of Paper 1.

The key difference between Lambertian scattering and diffusive scattering is the change of the measure. For diffusive scattering, the measure is $\sin(\theta)d\theta d\phi$ whereas for Lambertian scattering the measure is $\cos(\theta)\sin(\theta)d\theta d\phi$. Thus, since the geometric path length is $\sqrt{2R^2 - 2R^2 \cos(\pi - 2\theta)} = 2R \cos(\theta)$ (where θ is the angle measured from the normal), we have that the mean free path, l , is given by the following:

$$l = \left[(2R)(2\pi) \int_0^{\pi/2} \cos^2(\theta) \sin(\theta) d\theta \right] \left[(2\pi) \int_0^{\pi/2} \sin(\theta) \cos(\theta) d\theta \right]^{-1} = (4R/3)$$

In the second quantity above, the second bracket (which has a value of π) corresponds to normalization by the total number of electrons. Thus, we have shown that for a sphere we have $l = 4R/3$, which is precisely **Equation 17** of

Paper 1.

Section 2: Lambertian surface scattering model for nanoshells

For nanoshells, we will have two contributions to the mean free path. Electrons that start from the inner sphere will have a mean free path l_1 . Electrons that start from the outer surface will have mean free path l_2 . These electrons will either end on the outer surface (for large angles) or on the inner surface (for smaller angles). The average mean free path will be given by:

$$\frac{r^2 l_1 + R^2 l_2}{r^2 + R^2},$$

where r and R are the inner and outer sphere radii, respectively. In the below, we introduce the quantity $\alpha \equiv r/R$ for convenience.

Scattering from the inner surface to the outer surface

In this case, the path length may be written as a sum of two integrals (see <https://hackmd.io/@aligho/BkLmTZFubx> (<https://hackmd.io/@aligho/BkLmTZFubx>)). Thus, we write $l_1 = l_1^1 + l_1^2$. The first integral, which gives us l_1^1 , is given by the following:

$$\begin{aligned} l_1^1 &= \frac{2\pi}{\pi} \int_0^{\pi/2} R \sin(\theta) \cos(\theta) \sqrt{1 - \alpha^2 \sin^2(\theta)} d\theta = \frac{1}{\alpha^2} \int_{1-\alpha^2}^1 R \sqrt{u} du \\ &= \frac{2R}{3\alpha^2} \left[1 - (1 - \alpha^2)^{3/2} \right], \end{aligned}$$

where we have made a judicious change of variables by taking:

$$u = 1 - \alpha^2 \sin^2(\theta) \rightarrow du = -2\alpha^2 \sin(\theta) \cos(\theta)$$

We may just as easily calculate l_1^2 as follows:

$$l_1^2 = -2R\alpha \int_0^{\pi/2} \sin(\theta) \cos^2(\theta) d\theta = -\frac{2R\alpha}{3}$$

Thus, we have that the mean free path for electrons starting on the inner surface is given by the following:

$$l_1 = l_1^1 + l_1^2 = \frac{2R}{3\alpha^2} \left[1 - (1 - \alpha^2)^{3/2} - \alpha^3 \right]$$

Scattering from outer surface to outer surface

We now move on to the calculation of l_2 . We write $l_2 = l_2^1 + l_2^2$, where l_2^1 corresponds to the contribution from electrons scattering from the outer surface to the outer surface and, similarly, l_2^2 corresponds to electrons scattering from the outer surface to the inner surface. For electrons scattering from the outer surface to the outer surface, we find:

$$l_2^1 = 4R \int_{\arcsin(\alpha)}^{\pi/2} \sin(\theta) \cos^2(\theta) d\theta = \frac{4R}{3} \left[(1 - \alpha^2)^{3/2} \right]$$

Scattering from outer surface to inner surface

As in <https://hackmd.io/@aligho/BkLmTZFubX> (<https://hackmd.io/@aligho/BkLmTZFubX>), l_2^2 is the sum of two integrals. Thus, we write $l_2^2 = l_2^{2,1} + l_2^{2,2}$, where $l_2^{2,1}$ is given by the following:

$$l_2^{2,1} = 2R \int_0^{\arcsin(\alpha)} \sin(\theta) \cos^2(\theta) d\theta = \frac{2R}{3} \left[1 - (1 - \alpha^2)^{3/2} \right]$$

Similarly, $l_2^{2,2}$ is given by the following:

$$l_2^{2,2} = -2r \int_0^{\arcsin(\alpha)} \sin(\theta) \cos(\theta) \sqrt{1 - (1/\alpha^2) \sin^2(\theta)} d\theta = -r\alpha^2 \int_0^1 \sqrt{u} du = -\frac{2r\alpha^2}{3},$$

where we have made a change of variables:

$$u = 1 - (1/\alpha^2) \sin^2(\theta) \rightarrow du = -(2/\alpha^2) \sin(\theta) \cos(\theta) d\theta$$

Thus, we find that the average mean free path for particles starting on the outer surface is is given by the following:

$$l_2^{2,1} + l_2^{2,2} + l_2^1 = \frac{2R}{3} \left[1 - (1 - \alpha^2)^{3/2} - \alpha^3 + 2(1 - \alpha^2)^{3/2} \right]$$

Thus, the total average mean free path is given by:

$$\begin{aligned} l &= \frac{r^2 l_1 + R^2 l_2}{r^2 + R^2} = \frac{4R^3}{3(r^2 + R^2)} \left[1 - (1 - \alpha^2)^{3/2} - \alpha^3 \right] + \frac{4R^3}{3(r^2 + R^2)} (1 - \alpha^2)^{3/2} \\ &= \frac{4}{3} \frac{R^3(1 - \alpha^3)}{(r^2 + R^2)} = \frac{4}{3} \frac{(R^3 - r^3)}{(R^2 + r^2)}, \end{aligned}$$

which is **Equation 14** of **Paper 2**, as desired.